

Find the gradient of a line segment, the midpoint of the line interval and the distance between 2 distinct points on the Cartesian plane

Learning intention

- find the gradient of a line segment, the midpoint of the line interval and the distance between 2 distinct points on the Cartesian plane.

Teach and model

- using dynamic graphing software and superimposed images.
- investigating graphical and algebraic techniques for finding the midpoint and gradient of the line segment between 2 points.
- investigating how coordinate geometry and aspects of linear algebra play a fundamental role in machine learning and predictive algorithms.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.